PROJECT REPORT

On

Loan Approval Prediction using Machine Learning

Submitted in Partial Fulfillment of Award of

BACHELOR OF TECHNOLOGY

In

Computer Science and Engineering

By

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ALLIANCE SCHOOL OF ADVANCED COMPUTING

ALLIANCE UNIVERSITY

BENGALURU

April 2026

ALLIANCE SCHOOL OF ADVANCED COMPUTING

BACHELOR OF TECHNOLOGY

COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the project work entitled "**Loan approval prediction**" **2411021060110, Middinti Madhusudhan** in partial fulfilment for the award of the degree of Bachelor of Technology degree of Bachelor of Technology (CSE) of Alliance University, is a Bonafide work accomplished under our supervision and guidance during the academic year 2025-2026. This thesis report embodies the results of original work and studies conducted by students and the contents do not form the basis for the award of any other degree to the candidate or anybody else.

Dr R C Karpagalakshmi
Professor / CSE
(Supervisor)

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DECLARATION

I/We hereby declare that the project entitled "**Loan Approval Prediction**"submitted by me in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (CSE) of ASAC, Alliance University, is a record of my/our work carried under the supervision and guidance of Dr RC Karpagalakshmi

We confirm that this report truly represents the work undertaken as a part of our project work. This work is not a replication of work done previously by any other person. We also confirm that the contents of the report and the views contained therein have been discussed and deliberated with the faculty guide.

Name of the Student	University Registration Number	Signature
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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who supported and guided me throughout the completion of this project. This second year project has been a significant learning experience, and I am deeply thankful to everyone who contributed to its success.

First and foremost, I extend my heartfelt thanks to my project guide, **Dr R C Karpagalakshmi**
Professor / CSE , for her invaluable guidance, constant support, and insightful feedback at every stage of this project. Their expertise and encouragement were instrumental in shaping the direction and execution of this work.

I am also grateful to the **Dr V Vivek , Deputy Director**, for his leadership and for fostering an academic environment that encourages innovation and learning.

Additionally, I would like to thank the Head of Department, Program Co- ordinators and all the faculty members of the **CSE** department at **Alliance University** for providing a supportive learning environment and necessary resources.

A special note of thanks to my teammates and friends for their collaboration, ideas, and motivation during challenging times. I also appreciate the support and encouragement from my family, who have always been a source of strength and inspiration.

Lastly, I would like to acknowledge all the authors and researchers whose work I referred to while carrying out this project.

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1. Abstract

This project focuses on predicting whether a customer will repay a loan using machine learning techniques. The system analyzes customer financial and credit data to make accurate predictions. Logistic Regression is used as the primary model, achieving an accuracy of approximately 88%. This helps in reducing risk and improving decision-making in financial institutions.

2. Introduction

Loan approval and repayment prediction are critical tasks in the banking sector. Traditional methods rely on manual analysis, which is time-consuming and prone to errors. Machine learning provides an efficient way to automate this process by learning patterns from historical data and making predictions.

3. Problem Statement

The objective of this project is to predict whether a customer will repay the loan based on various features such as income, credit score, and financial history.

4. Objectives

- To analyze loan dataset
- To preprocess and clean data
- To build a prediction model
- To evaluate model performance
- To predict loan repayment

5. Dataset Description

The dataset contains customer-related information such as:

- Age
- Gender
- Annual Income
- Credit Score
- Loan Amount
- Employment Status

The target variable is:

- **loan_paid_back** (1 = Yes, 0 = No)

6. Methodology

- **Data Preprocessing**
  - Missing values handled using mean and mode
  - Categorical values converted into numerical format
- **Feature Engineering**
  - Created a new feature: income_ratio
- **Model Selection**
  - Logistic Regression used for classification

7. Implementation

The implementation is done using Python with libraries such as:

- Pandas
- NumPy
- Scikit-learn

Steps include:

- Loading dataset
- Cleaning data
- Splitting data
- Training model
- Making predictions

8. Results and Evaluation

The model achieved an accuracy of approximately **88.35%**, indicating good performance. The system successfully predicts whether a customer will repay the loan.

9. Advantages and Disadvantages

Advantages:

- Fast prediction
- Reduces human error
- Data-driven decisions

Disadvantages:

- Not 100% accurate
- Depends on data quality
- Limited features

10. Applications

- Banking sector
- Loan approval systems
- Credit risk analysis
- Financial institutions

11. Future Scope

- Use advanced models like Random Forest
- Improve feature engineering
- Deploy as a web application

12. Conclusion

This project demonstrates the effectiveness of machine learning in predicting loan repayment. It improves efficiency, reduces risk, and supports better financial decisions.

13. References

- Scikit-learn documentation
- Python documentation
- Machine learning tutorials